An important aspect of a mixing system is the *decay of autocorrelation* [338]. Note that

$$\begin{aligned}\int_{\Sigma_{\mathrm{E}}} \phi(z)\rho(z,t)\mathrm{d}\omega &= \int_{\Sigma_{\mathrm{E}}} \phi(z)e^{t\mathcal{L}^{\dagger}}\rho(z,0)\mathrm{d}\omega \\ &= \int_{\Sigma_{\mathrm{E}}} \left[e^{t\mathcal{L}}\phi(z)\right]\rho(z,0)\mathrm{d}\omega \\ &= \int_{\Sigma_{\mathrm{E}}} \phi(\mathscr{F}_t(z))\rho(z,0)\mathrm{d}\omega \\ &= \int_{\Sigma_{\mathrm{E}}} \phi(\mathscr{F}_t(z))\frac{\rho(z,0)}{\rho_{\mathrm{E}}(z)}\rho_{\mathrm{E}}(z)\mathrm{d}\omega. \end{aligned} \tag{5.10}$$

We assume $\psi(z) \overset{\text{def}}{=} \frac{\rho(z,0)}{\rho_{\mathrm{E}}(z)}$ is bounded on Σ_{E}, and moreover

$$\mathrm{Av}_{H=\mathrm{E}}(\psi) = \int_{\Sigma_{\mathrm{E}}} \frac{\rho(z,0)}{\rho_{\mathrm{E}}(z)}\rho_{\mathrm{E}}(z)\mathrm{d}\omega = 1$$

(i.e., the initial density is a probability density on the energy surface), and that the system is mixing. Then the integral (5.10) is well defined and we may view it as a two-point correlation function of ϕ and ψ. By the mixing property it must converge to $\int_{\Sigma_{\mathrm{E}}} \phi(z)\rho_{\mathrm{E}}\mathrm{d}\omega$, thus the average value of ϕ is ultimately independent of the initial conditions. We have

$$\lim_{t\to\infty} \int_{\Sigma_{\mathrm{E}}} \phi(\mathscr{F}_t(z))(\psi(z)-1)\rho_{\mathrm{E}}\mathrm{d}\omega = 0$$

We say that the system loses memory of its initial state. More generally, in a mixing system, if $\boldsymbol{a}$ and $\boldsymbol{b}$ are two arbitrary functions of the phase variables, then

$$\lim_{t\to\infty} C_{\boldsymbol{ab}}(t) = \mathrm{Av}_{H=\mathrm{E}}(\boldsymbol{a}) \cdot \mathrm{Av}_{H=\mathrm{E}}(\boldsymbol{b}).$$

So if either property has mean zero, then the absolute value of the autocorrelation function decays with time to zero. In practice, it may take the system a very long time to decorrelate from its initial state [135].